

BARDUFOSS, Norway

WMO: 010230

Lat: 69.056N

Lon: 18.540E

Elev: 77

StdP: 100.41

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.2	-24.0	-29.0	0.3	-26.1	-26.5	0.3	-23.8	11.0	-3.7	9.7	-2.1	0.8	290	0.700

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	24.0	15.3	21.9	14.5	19.8	13.6	16.5	21.7	15.3	20.0	14.3	18.6	2.9	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.2	10.2	18.5	13.1	9.5	17.0	12.1	8.9	16.0	46.5	21.6	43.0	20.1	40.2	18.7	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.1	6.1	-28.3	26.8	2.6	2.2	-30.2	28.4	-31.7	29.7	-33.1	31.0	-35.0	32.6		
(5)			-28.5	18.1	2.5	1.3	-30.3	19.0	-31.8	19.8	-33.2	20.5	-35.1	21.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.5	-9.3	-8.5	-4.9	0.5	5.8	10.4	13.7	12.1	7.8	1.3	-4.7	-7.1	
	DBStd	9.53	7.62	7.09	5.59	3.95	3.60	3.47	3.33	3.02	3.41	4.69	6.11	7.30	
	HDD10.0	3378	598	518	463	285	138	36	5	12	83	270	441	529	
	HDD18.3	6157	856	751	721	535	387	239	149	195	317	528	691	787	
	CDD10.0	268	0	0	0	0	8	48	118	77	16	1	0	0	
	CDD18.3	6	0	0	0	0	0	1	4	1	0	0	0	0	
	CDH23.3	87	0	0	0	0	3	9	55	19	0	0	0	0	
	CDH26.7	11	0	0	0	0	0	1	9	2	0	0	0	0	
Wind	WSAvg	2.2	1.9	2.1	2.4	2.6	2.8	2.7	2.4	2.0	2.0	1.8	1.7	2.0	
Precipitation	PrecAvg	865	86	74	74	46	44	49	71	69	86	100	72	94	
	PrecMax	1077	211	145	201	125	97	95	164	148	188	150	148	169	
	PrecMin	678	11	9	17	7	10	4	18	21	30	35	3	15	
	PrecStd	103	54	30	49	28	23	22	34	28	43	35	43	44	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.4	6.1	8.2	13.2	22.7	24.7	27.6	25.9	19.8	14.0	8.8	7.3	
		MCWB	3.7	3.5	3.4	7.0	12.7	14.4	18.1	15.6	13.2	9.5	6.4	4.7	
	2%	DB	4.8	4.5	5.9	10.8	18.5	22.0	24.6	22.7	17.3	11.1	6.7	5.7	
		MCWB	2.7	2.4	2.7	5.6	10.5	13.3	16.0	15.3	12.3	8.0	4.7	3.5	
	5%	DB	3.0	3.2	4.4	8.2	14.9	19.1	22.4	20.0	15.5	9.5	5.0	4.1	
		MCWB	1.5	1.5	1.8	4.3	8.4	12.0	15.2	14.4	11.4	7.2	3.2	2.1	
	10%	DB	1.1	1.3	3.0	6.4	12.3	17.0	20.4	17.9	13.7	8.0	3.2	2.5	
		MCWB	-0.1	-0.1	0.8	3.1	7.1	11.1	14.4	13.3	10.4	6.0	1.6	0.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	4.0	4.2	7.5	13.0	15.7	18.9	17.8	14.4	9.8	6.7	5.1
			MCDB	5.9	5.5	6.8	12.2	22.0	22.2	26.3	22.7	18.0	12.5	8.2	6.5
2%		WB	2.7	2.6	3.1	6.0	10.6	14.1	17.2	15.9	12.9	8.6	5.0	3.7	
		MCDB	4.3	4.2	5.3	9.8	17.4	20.3	22.8	20.6	16.4	10.6	6.4	5.5	
5%		WB	1.5	1.6	2.1	4.6	8.9	12.8	15.9	14.8	11.8	7.3	3.3	2.3	
		MCDB	2.8	3.1	4.1	7.7	14.1	18.3	20.8	18.9	14.8	9.1	4.8	3.9	
10%		WB	0.0	0.1	0.9	3.4	7.5	11.6	14.9	13.8	10.7	6.1	1.8	1.0	
		MCDB	1.1	1.2	2.7	6.0	11.6	16.0	19.4	17.6	13.3	7.8	3.0	2.2	
Mean Daily Temperature Range		5% DB	MDBR	8.0	8.5	9.5	8.4	7.7	7.0	7.9	8.3	7.9	6.4	6.8	7.8
			MCDBR	7.5	7.2	7.2	10.5	13.4	11.6	12.7	13.2	11.5	7.4	7.6	7.4
	MCWBR		6.2	6.0	5.2	6.4	7.0	5.3	5.9	6.9	7.0	5.2	5.9	5.8	
	5% WB	MCDBR	7.1	7.0	6.7	9.4	12.7	10.3	11.4	11.4	9.9	7.0	7.1	7.1	
		MCWBR	6.0	6.1	5.1	6.0	6.8	5.1	5.8	6.4	6.4	5.3	5.7	5.8	
Clear-Sky Solar Irradiance	taub	0.210	0.263	0.309	0.331	0.343	0.344	0.349	0.336	0.314	0.274	0.361	N/A		
	taud	1.914	2.054	2.099	2.183	2.318	2.441	2.474	2.494	2.521	2.210	2.634	N/A		
	Ebn at Noon	33	474	687	801	841	854	836	805	727	469	65	0		
	Edh at Noon	5	45	83	104	103	94	89	77	58	38	9	0		
All-Sky Solar Radiation	RadAvg	0.02	0.33	1.55	3.45	4.77	4.83	4.43	3.17	1.67	0.52	0.06	0.00		
	RadStd	0.00	0.06	0.18	0.25	0.31	0.55	0.44	0.27	0.17	0.08	0.01	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page